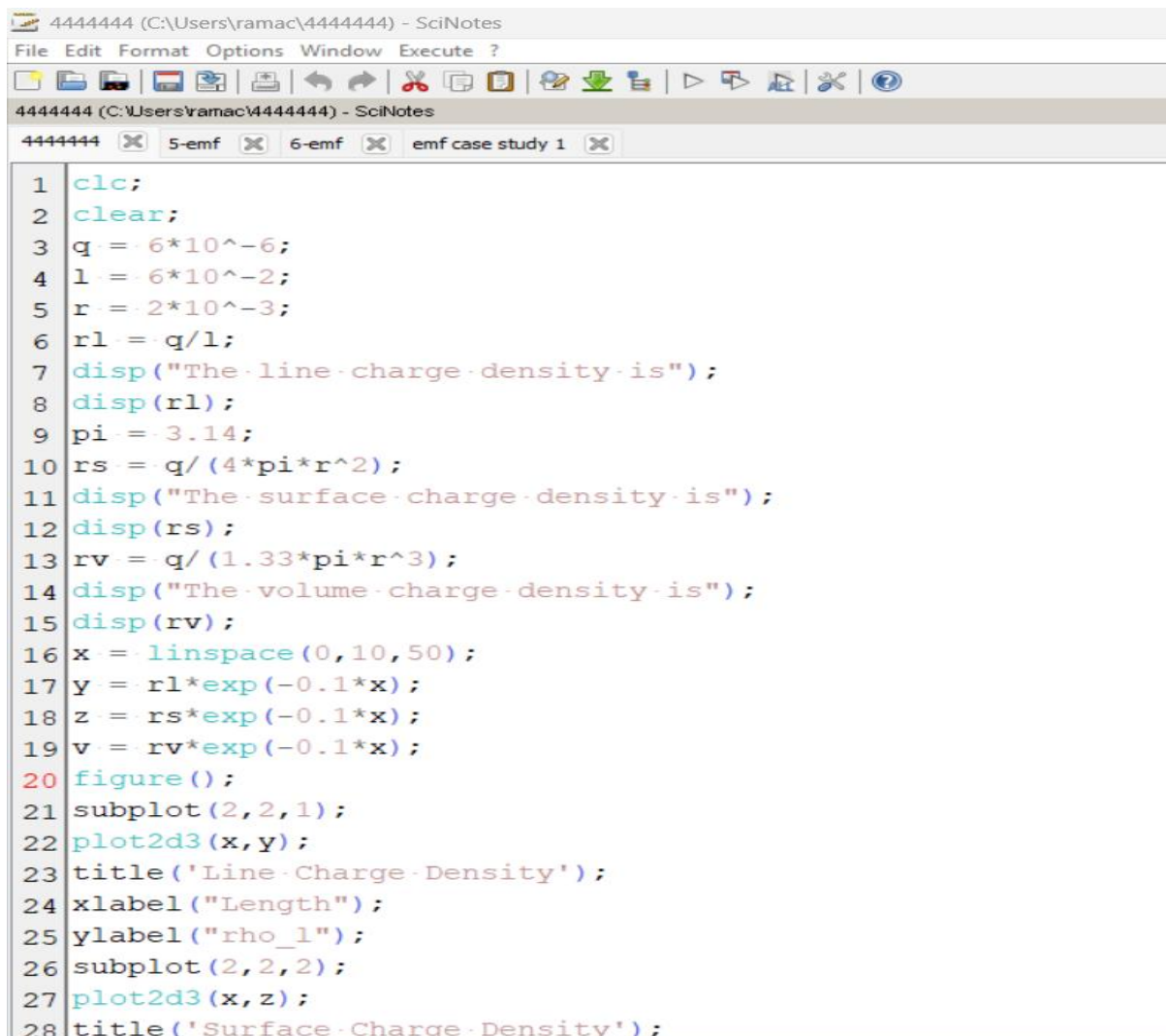


Exp. No: 4 Analysis of Continuous Charge Distribution in electric field. To find the (i) line charge density, (ii) Surface charge density and (iii) Volume charge density for any specific (a) line, (b) surface or(c) volume.

Aim: To write program for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB.

Software required: SCILAB version 6.0.1

Practical output:



```
4444444 (C:\Users\ramac\4444444) - SciNotes
File Edit Format Options Window Execute ?
4444444 (C:\Users\ramac\4444444) - SciNotes
4444444 5-emf 6-emf emf case study 1
1 clc;
2 clear;
3 q = -6*10^-6;
4 l = -6*10^-2;
5 r = -2*10^-3;
6 rl = q/l;
7 disp("The line charge density is");
8 disp(rl);
9 pi = 3.14;
10 rs = q/(4*pi*r^2);
11 disp("The surface charge density is");
12 disp(rs);
13 rv = q/(1.33*pi*r^3);
14 disp("The volume charge density is");
15 disp(rv);
16 x = linspace(0,10,50);
17 y = rl*exp(-0.1*x);
18 z = rs*exp(-0.1*x);
19 v = rv*exp(-0.1*x);
20 figure();
21 subplot(2,2,1);
22 plot2d3(x,y);
23 title('Line Charge Density');
24 xlabel("Length");
25 ylabel("rho_l");
26 subplot(2,2,2);
27 plot2d3(x,z);
28 title('Surface Charge Density');
```

```
Scilab 6.1.1 Console
File Edit Control Applications ?
[Icons]
Scilab 6.1.1 Console

"The line charge density is"

0.0001

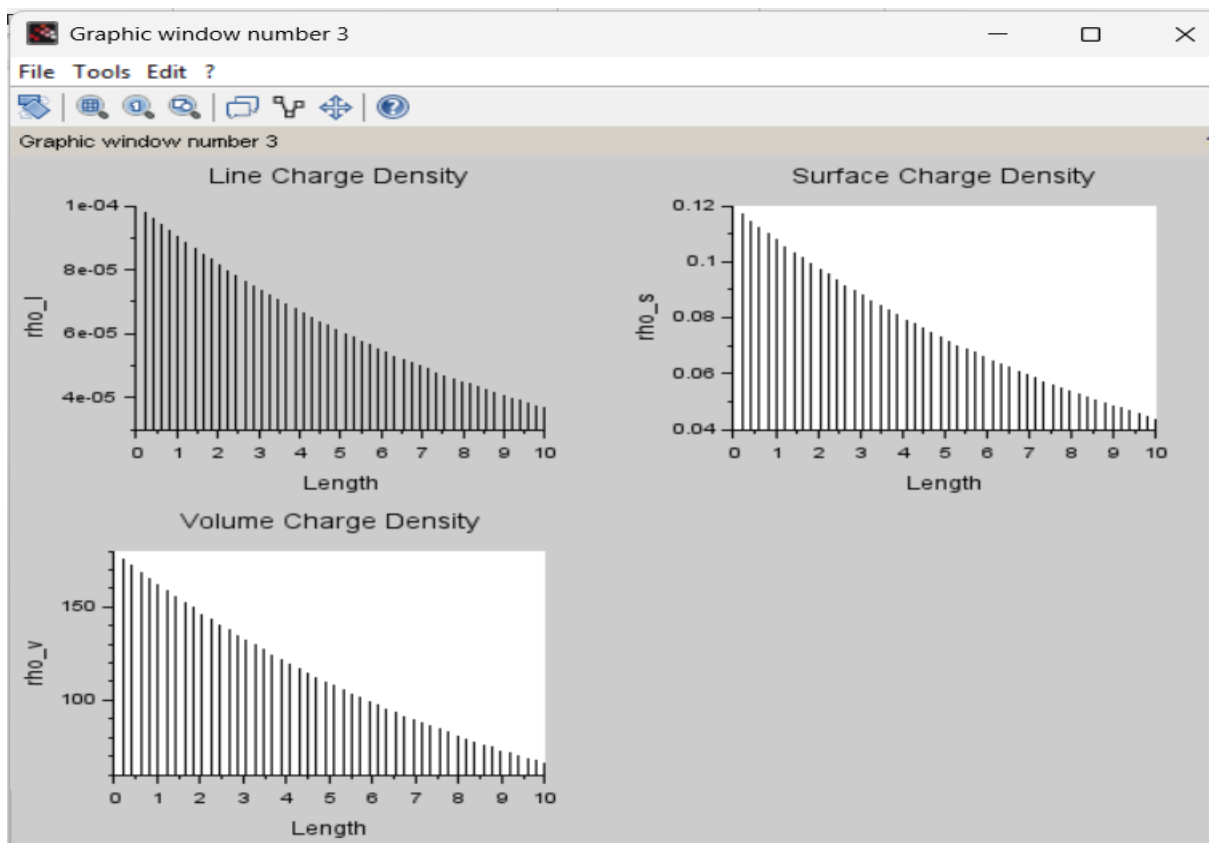
"The surface charge density is"

0.1194268

"The volume charge density is"

179.58910

-->
```



Theoretical Output:

④

$$\begin{aligned}\text{charge } (q) &= 6 \times 10^{-6} \\ \text{length } (l) &= 6 \text{ cm} \\ \text{radius } (r) &= 2 \text{ mm}\end{aligned}$$
$$\rho_l = \frac{q}{l} \Rightarrow \frac{6 \times 10^{-6}}{6 \times 10^{-2}} \Rightarrow 1 \times 10^{-4} \text{ C/m}$$
$$\rho_s = \frac{q}{4\pi r^2} \Rightarrow \frac{6 \times 10^{-6}}{4 \times 3.14 \times (2 \times 10^{-3})^2} \Rightarrow 0.1194 \text{ C/m}^2$$
$$\rho_v = \frac{q}{1.33 \times 10^{-3}} \Rightarrow \frac{6 \times 10^{-6}}{1.33 \times 3.14 \times (2 \times 10^{-3})^3} \Rightarrow 179.69 \text{ C/m}^3$$

0/2

Result : Thus the program was verified for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB was successfully.